**1** **(a)** 21 **[B2]**

M1 for 13 or 8

**(b)** 80 **[B2]**

M1 for 16 or 5

*[Total Marks: 4]*

**2** **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

**(b)** $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

**(c)** $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

*[Total Marks: 7]*

**3** **(a)** 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

**(b)** 5.399 **[B1]**

**(c)** 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

*[Total Marks: 5]*

**4** **(a)** 14 **[B1]**

**(b)** 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

**(c)** 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

*[Total Marks: 5]*

**5** **(a)** **(i)** $1.49315\ldots$ **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

**(ii)** 1.49 **[B1]**

**(b)** 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

*[Total Marks: 5]*

**6** **(a)** 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

**(b)** $-2$ **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or $-4$ **[M1]**

**(c)** 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

*[Total Marks: 8]*